

IL-23R : IL-23p19

Blocking a clinically validated cytokine–receptor interaction with an orally dosed small molecule: structural basis, ligandable sites, discovery routes, and a preclinical target product profile for plaque psoriasis.

Interaction IL-23p19 (IL23A) · IL-23R site 3	Indication Moderate-to-severe plaque psoriasis	Modality Oral small molecule (Ro5 / bRo5)
Validation Human genetics + 6 approved drugs	Precedent Icotrokinra, oral peptide, FDA 2026	Stage framed Hit ID → candidate selection

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Six figures and nine tables are distributed through sections 2 to 9. The target product profile is Table 7 in section 8. Sourcing conventions and the boundary between cited fact and the author’s own proposals are stated in the note at the end of section 1 and in the colophon on the final page.

What the opportunity actually is, now that an oral drug exists

The IL-23 pathway is one of the most thoroughly validated targets in immunology, and as of March 2026 the specific interaction described in this dossier — IL-23p19 docking onto the N-terminal Ig domain of IL-23R — has been drugged orally in humans. Icotrokinra, a 13-residue macrocyclic peptide, is approved for plaque psoriasis. That removes essentially all target risk from a small-molecule programme and replaces it with a sharply defined competitive question.

The mechanistic argument for a small molecule is no longer "can this interaction be blocked orally". It is that the approved peptide reaches the clinic despite an oral bioavailability of roughly 0.1–0.3% in animals and under 1% overall, requires 200 mg once daily taken fasted 30 minutes before food, and delivers a plasma $C_{\max}$ of about 3.6 ng/mL. Efficacy is carried entirely by picomolar potency (K_D 7.1 pM against the human IL-23R extracellular domain; 20–30 pM in cellular assays), not by exposure. A conventional small molecule with 30–50% bioavailability, no food effect, and modest tissue distribution would sit in a different pharmacological regime, and would open dose forms, combinations, and tissue compartments — psoriatic skin, gut mucosa, joint — that a peptide with sub-1% absorption cannot easily reach.

The counter-argument is equally sharp. This is a shallow, extended protein–protein interface of the classic "site 3" cytokine type, with no natural small-molecule ligand and no endogenous pocket. The one credible small-molecule-sized pocket published to date is **cryptic**: it does not exist in apo IL-23 and only forms when an inhibitory peptide binds, at a cost in conformational reorganisation that a 400-dalton compound must pay for out of its own binding energy. A programme here is a hard PPI programme with a de-risked biology, not an easy programme.

The five conclusions this dossier supports

- 1. Target validation is complete and does not need further work.** Human loss-of-function genetics (IL23R R381Q), four approved anti-p19 antibodies, one anti-p40 antibody, and now an oral IL-23R peptide antagonist all converge.
- 2. Structural coverage is good but incomplete in the one place it matters.** Ten-plus public structures define the quaternary complex and the site-3 hotspot at atomic resolution, but there is still no public co-structure of a non-peptidic ligand with either partner.
- 3. The best-supported small-molecule site is the induced-fit pocket on p19** revealed by peptide 23–446 (PDB 8UUI), which sits adjacent to and sterically occludes the IL-23R footprint. A second, non-overlapping pocket on IL-23 (PDB 6UIB) and a receptor-side D1 site mapped by HDX-MS give two further starting points.
- 4. Discovery must be built around cryptic-pocket methods** — mixed-solvent molecular dynamics, peptide-anchored structure-based design, and a fluorescence-polarisation assay run against a conformation-selective probe — because ensemble docking into the apo structure will fail.
- 5. Efficacy bar is set by icotrokinra, not by placebo.** Any candidate must credibly project toward IGA 0/1 near 65–70% and PASI 90 near 55% at week 16, with a placebo-like safety profile. A "me-too oral" that lands between apremilast and deucravacitinib will not be commercially viable.

How to read this document. Facts drawn from the published literature are cited by number to the reference list in §10. Numerical proposals in the target product profile (§8) are the author's construction for a hypothetical programme; they are engineering targets, not literature values, and are labelled as such. Nothing here constitutes clinical or investment advice.

A cytokine that only exists to make Th17 cells dangerous

IL-23 is a disulfide-linked heterodimer of a four-helix bundle, p19 (gene **IL23A**), and a soluble class-I cytokine receptor domain, p40 (gene **IL12B**), joined by an intersubunit disulfide between **Cys73** of p19 and **Cys199** of p40. p40 is shared with IL-12, where it pairs with p35; p19 is unique to IL-23. This asymmetry is the whole basis of selective targeting — anything that engages p19, or the receptor chain that reads p19, spares IL-12 and the IFN- γ arm of host defence by construction.

Receptor assembly is sequential and cooperative. IL-23R engages the tip of p19 through its N-terminal Ig-like domain alone; that engagement partially restructures p19 and constrains p40 so that the shared chain IL-12R β 1 can be recruited with high affinity. The assembled complex apposes JAK2 (on IL-23R) and TYK2 (on IL-12R β 1), driving STAT3 phosphorylation and, downstream, ROR γ t-dependent transcription of IL-17A, IL-17F, IL-22 and IL-19 in Th17 cells, $\gamma\delta$ T cells, ILC3s, Tc17 cells and tissue-resident memory T cells in skin.

In psoriasis this circuit is the disease. IL-23 from dermal dendritic cells maintains a pathogenic resident T-cell population in plaques; the IL-17 output drives keratinocyte hyperproliferation and the neutrophil-rich, β -defensin-2-high inflammatory signature that defines the lesion. Serum IL-17A, IL-17F, IL-22, IL-19 and BD-2 all fall on IL-23 pathway blockade and serve as usable pharmacodynamic markers.

The human genetic argument

The **IL23R** missense variant R381Q (rs11209026), carried by roughly 7% of chromosomes in European-ancestry populations, protects against psoriasis, Crohn's disease, ulcerative colitis, ankylosing spondylitis and asthma. Functionally it is hypomorphic rather than null: carriers show reduced IL-23-induced STAT3 phosphorylation and reduced IL-17A output from Th17 cells, with normal Th17 differentiation and preserved receptor expression and trafficking. This is the ideal genetic template for a drug — a partial, lifelong reduction in signalling that produces protection without an immunodeficiency phenotype, which is precisely what a reversible antagonist with an imperfect trough concentration will reproduce.

Table 1. Validation evidence for the IL-23p19 : IL-23R interaction, ordered from most to least direct for a small-molecule antagonist.

Evidence class	Observation	What it establishes	Residual gap
Oral chemical matter in humans	Ichotrokinra, a 13-mer macrocyclic peptide binding IL-23R, approved March 2026 for moderate-to-severe plaque psoriasis in adults and adolescents ≥12 y, ≥40 kg	Blocking this exact interface with an orally dosed synthetic molecule produces biologic-level skin clearance	Peptide, not small molecule; F <1%; fasted dosing
Antibody pharmacology	Guselkumab, risankizumab, tildrakizumab (psoriasis) and mirikizumab (IBD) neutralise p19; ustekinumab neutralises shared p40	Ligand neutralisation is efficacious and durable across IMiDs; p19-selective agents match or beat p40 blockade	Parenteral; does not speak to oral small-molecule feasibility
Human genetics	IL23R R381Q hypomorph, minor allele ≈7%, protective across psoriasis, IBD, AS, asthma	Partial, chronic pathway reduction is protective and well tolerated over a lifetime	Variant acts on the cytoplasmic tail/JAK2 coupling, not the ligand interface
Head-to-head clinical	Ichotrokinra 200 mg QD superior to deucravacitinib 6 mg QD and placebo at week 16 (ICONIC-ADVANCE 1 & 2)	Direct pathway blockade beats upstream kinase modulation on efficacy in the same oral setting	Sets a high, not a permissive, efficacy bar for followers
Class safety	Long-term p19 inhibitor reviews report no elevated serious infection, opportunistic infection, TB reactivation, oral candidiasis, NMSC or MACE signal	On-target chronic IL-23 blockade is tolerated; the safety burden shifts to off-target chemistry	Small-molecule-specific liabilities (CYP, hERG, reactive metabolites) are unaddressed by class data
Non-clinical pathway	IL-23-driven murine skin models (intradermal IL-23 ear model, imiquimod) reproduce IL-17/IL-22-dependent acanthosis	Usable in vivo pharmacology exists	Human-selective binders need surrogate tools or humanised systems

Where IL-23R is expressed, and why that matters for a small molecule

IL-23R is restricted: Th17 and Th17.1 cells, γδ T cells, ILC3s, NKT and NK subsets, Tc17 cells, some mast cells and a minority of myeloid cells. Receptor density on primary human cells is low — a fact that has shaped every assay in this field, since occupancy-based readouts are hard and function-based readouts (pSTAT3, IL-17 release) dominate. Two consequences for chemistry: first, a low receptor reserve means fractional occupancy translates efficiently into pathway inhibition, favouring the achievability of the potency targets in §8; second, target-mediated drug disposition is negligible, so pharmacokinetics will be governed by conventional ADME rather than by target sink.

Three interfaces, one of which is worth attacking

The quaternary IL-23 receptor complex was solved crystallographically in 2021 and follows the IL-6 family nomenclature of sites 1, 2 and 3, but with a non-canonical twist: IL-12R β 1 binds directly to the shared p40 subunit rather than to the helical cytokine. IL-23 acts as a molecular bridge, holding IL-23R and IL-12R β 1 close enough for JAK2/TYK2 trans-phosphorylation.

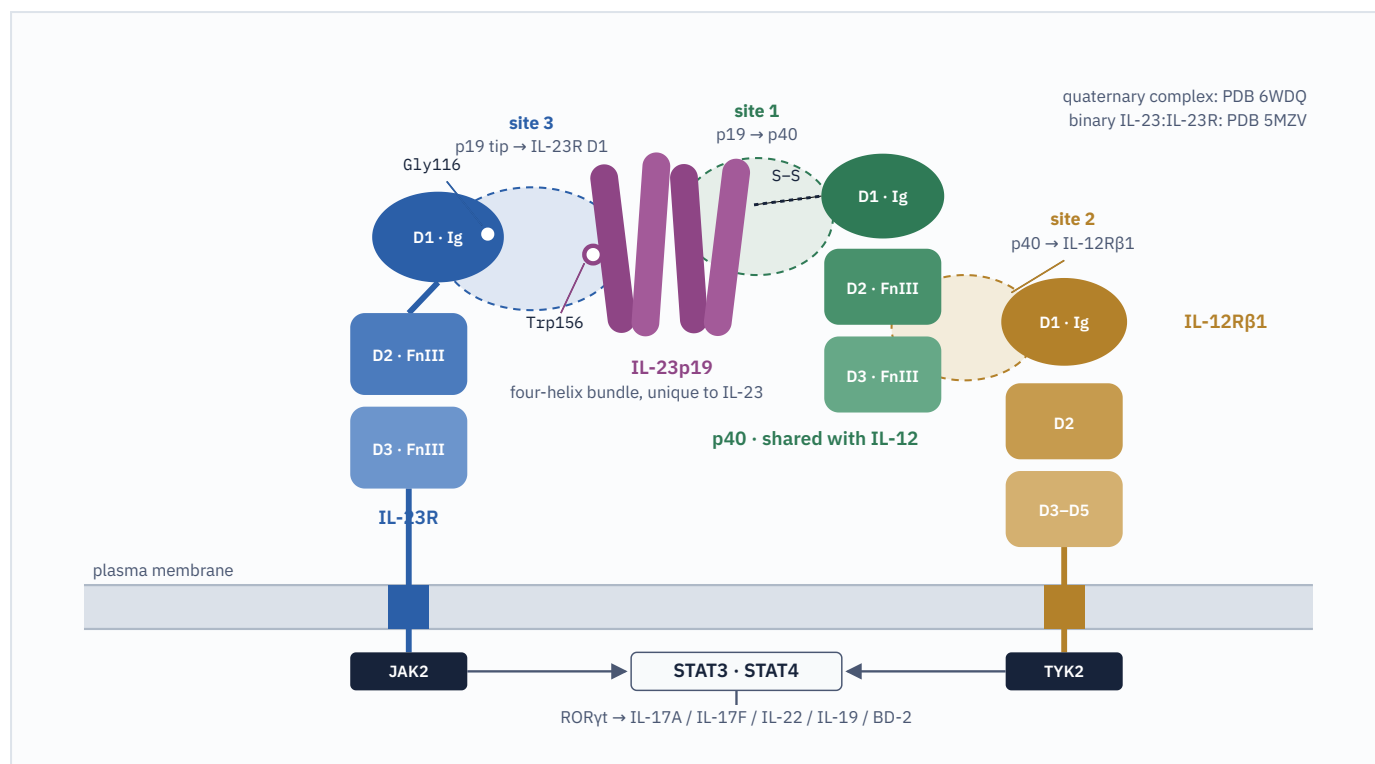


Figure 1. Architecture of the IL-23 receptor complex. IL-23R engages the tip of p19 exclusively through its N-terminal Ig-like domain (site 3), anchored by Trp156 of p19 packing against the inner β -sheet of IL-23R D1 near Gly116, with the receptor's N-terminal extension contacting the top of p19 helix D. p19 and p40 are covalently joined at site 1 (Cys73–Cys199). IL-12R β 1 binds the "back" of p40 at the D1/D2 junction (site 2) without contacting p19. Domain proportions are schematic. Sites and hotspot residues after refs 3–5.

Only site 3 is a legitimate target for a selective small molecule. Site 2 lies entirely on the shared p40 subunit and on the shared receptor chain, so blocking it would also block IL-12 and reintroduce the ustekinumab-like breadth that p19-selective agents were designed to escape. Site 1 buries an intersubunit disulfide and is formed co-translationally during heterodimer secretion, which makes it a poor target for a circulating antagonist. Site 3 is unique to IL-23 on both sides of the interface.

Table 2. Public structural resources. All entries are freely available from the RCSB PDB and constitute the working structural package for a programme.

PDB	Contents	Method	Why it matters for chemistry
Receptor complexes — define the interface to be blocked			
6WDQ	IL-23p19/p40 : IL-23R : IL-12Rβ1 quaternary complex	X-ray	The reference geometry. Source of the site-3 footprint used to judge whether any candidate pocket is orthosteric.
5MZV	IL-23 : IL-23R (D1) with nanobody 22E11	X-ray	Binary complex; IL-23R rotated ~18° relative to 6WDQ, giving a two-point conformational ensemble for the interface.
Free cytokine — the docking start point, and the trap			
5MXA	Apo human IL-23	X-ray	The conformation a naïve virtual screen would use. The orthosteric pocket is <i>absent</i> here.
3DUH	Human IL-23, 2.3 Å	X-ray	High-quality apo reference; defines the site-1 "arginine pocket" on p40.
3D87	Human IL-23	X-ray	Independent apo determination for ensemble construction.
Ligand-bound — the actual chemical starting points			
8UUI	IL-23 : inhibitory peptide 23–446	X-ray	The key structure. Reveals an induced-fit orthosteric pocket on p19 that occludes IL-23R binding. Use this conformation, not the apo one, for design.
6UIB	IL-23 : peptide 23–652	X-ray	A second, distal and non-overlapping pocket on IL-23 (probe affinity ≈19 nM). Serves as a counter-screen site and an alternative allosteric hypothesis.
3D85	IL-23 : neutralising Fab	X-ray	Antibody epitope on p19 overlapping the receptor site; a template for "antibody-inspired" pharmacophores.
5NJD	IL-23 : briakinumab Fab	X-ray	Second neutralising epitope; useful for mapping which surface patches tolerate bulky occupancy.
4OE8	IL-23 : Alphabody MA12 scaffold 40G9, 5MJ4	X-ray	Shows Trp156 inserting into an engineered groove, and an MA12 tyrosine inserting into a p19 pocket formed by the AB/BC loops and the N-terminus of helix D — a distinct, exploitable sub-pocket.

The residue-level picture on p19

Everything a medicinal chemist needs to reason about is concentrated in a small number of positions.

Figure 2 places them on the sequence.

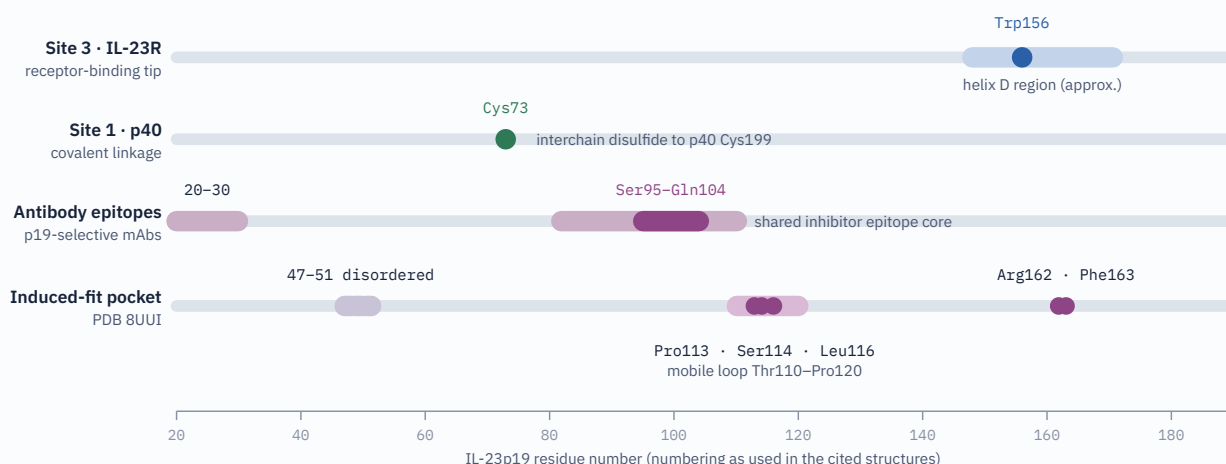


Figure 2. Functional map of IL-23p19. The site-3 anchor Trp156 sits in the helix-D region that IL-23R reads. The induced-fit pocket described in PDB 8UUI is lined by the mobile Thr110–Pro120 loop and by Arg162/Phe163; peptide binding also disorders residues 47–51. The Ser95–Gln104 stretch is the region of greatest epitope overlap among clinical p19 antibodies even though it is not itself a direct receptor contact. Band positions are drawn to the residue axis; where a source reports a region rather than individual contacts, the band is shaded lightly and marked approximate.

Table 3. Hotspot and pocket-lining residues, with the interaction each one makes and the design consequence.

Residue	Chain	Interaction reported	Design consequence
Trp156	p19	Indole packs against the inner β -sheet of IL-23R D1, near Gly116; the canonical site-3 aromatic anchor	Any ligand that immobilises or buries Trp156, or occupies the shallow groove it projects into, is mechanistically orthosteric. Highest-value single residue on the target.
Gly116	IL-23R	Receiving surface for the p19 tryptophan on the D1 β -sheet	Marks the receptor-side sub-pocket; the plausible landing zone for a receptor-binding chemotype.
Phe163	p19	π -stacking with the Phe14 and Tyr10 side chains of peptide 23–446	A ready-made aromatic box. Replace the peptide's two aromatics with one rigid biaryl or fused heteroaryl to recover entropy.
Arg162	p19	π -cation contact with the Trp18 indole of peptide 23–446	Cation– π donor: pair with an electron-rich heteroarene or an aryl carboxylate/tetrazole for a salt bridge as an alternative vector.
Ser114	p19	Backbone hydrogen bond to the Tyr19 backbone of the peptide	Backbone-mediated, so conserved across chemotypes; an amide or lactam NH is the natural mimic.
Leu116	p19	Backbone hydrogen bond to the Asn17 backbone of the peptide	Second backbone anchor; together with Ser114 defines a two-point recognition motif independent of side-chain chemistry.
Pro113	p19	CH– π contact with the Trp20 indole of the peptide	Small lipophilic shelf; good for a methyl or cyclopropyl vector rather than a large aromatic.
Thr110–Pro120	p19	Loop that shifts substantially on peptide binding, together with helices 2, 3 and 4	The conformational price of the pocket. Compounds must either pay it or be designed against the already-open 8UUI state.
47–51	p19	Becomes disordered upon peptide binding; the AB-loop segment is disordered in all published IL-23 structures	Entropically favourable disorder on binding — a rare positive for a PPI, but it makes crystallography of weak fragments harder.
Cys73 : Cys199	p19 : p40	Interchain disulfide at site 1	Confirms site 1 is not a viable antagonist site for a reversible circulating molecule.

A numbering trap worth knowing about before anyone opens a PDB file. p19 residue numbering is not consistent across the literature. The Alphabody paper explicitly notes that its numbering follows the UniProt sequence and therefore differs by 19 residues from the numbering used in earlier IL-23 depositions, which start from the mature protein. Any programme that mixes residue numbers taken from different papers — for example, a mutagenesis panel built from one source and a docking grid built from another — will silently mis-target by 19 positions. Fix a single convention at the start, state it in every document, and re-derive every residue number from the coordinates rather than from a figure legend.

Why the pocket is the whole story

The distinguishing feature of the 23–446 structure is that the binding site does not exist before the ligand arrives. Comparing apo IL-23 with the peptide complex shows the **Thr110–Pro120** loop displaced, helices 2, 3 and 4 shifted, and residues **47–51** released into disorder. The resulting cavity is offset from the direct receptor footprint, so inhibition works by steric occlusion of the site-3 approach rather than by direct competition for the exact contact atoms. Practically, this means three things: docking into **5MXA** will return nothing; fragment soaking into apo crystals will likely fail; and the assay used to find hits must be able to detect binding to a conformation that only a minority of molecules populate at any instant.

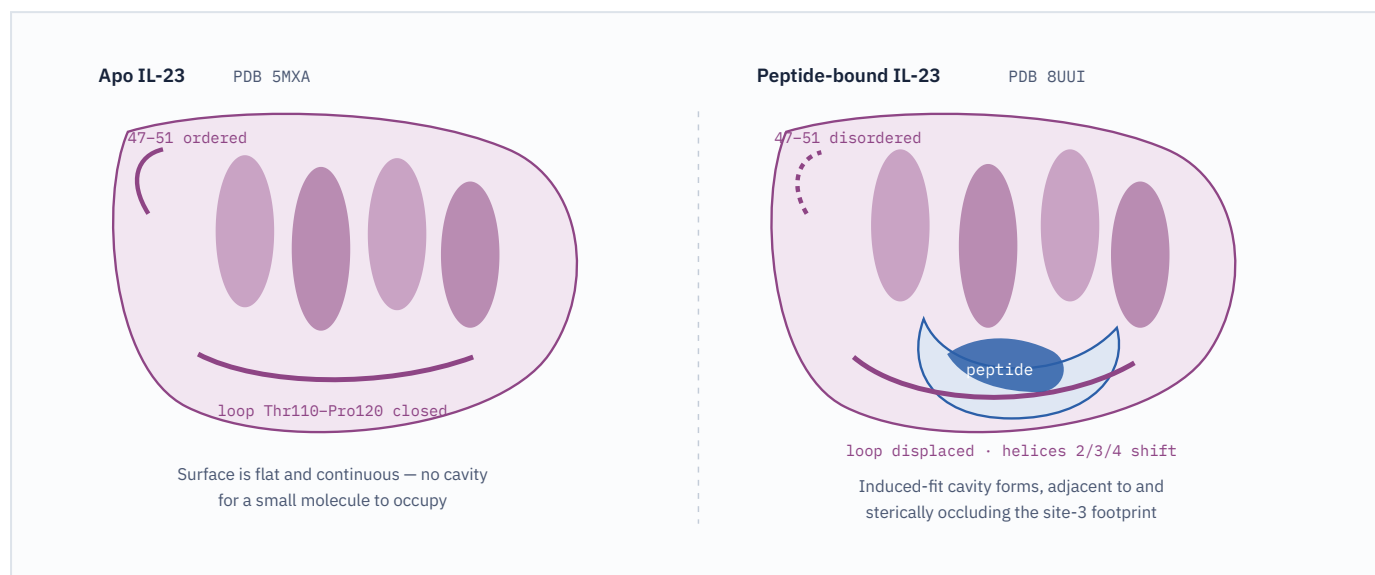


Figure 3. The pocket is cryptic. Schematic comparison of apo IL-23 with the 23–446 peptide complex, drawn from the conformational changes reported for these structures: displacement of the Thr110–Pro120 loop, shifts in helices 2, 3 and 4, and loss of order in residues 47–51. This is an abstraction of the reported changes, not a rendering of coordinates.

Five candidate sites, ranked honestly

A programme has to commit to a site before it commits to a screen. The published structural work defines five distinguishable places a small molecule could bind and interfere with IL-23 signalling. They are not equivalent: two are mechanistically sound and chemically hard, one is chemically attractive and mechanistically unproven, and two should be eliminated on selectivity or accessibility grounds before any money is spent.

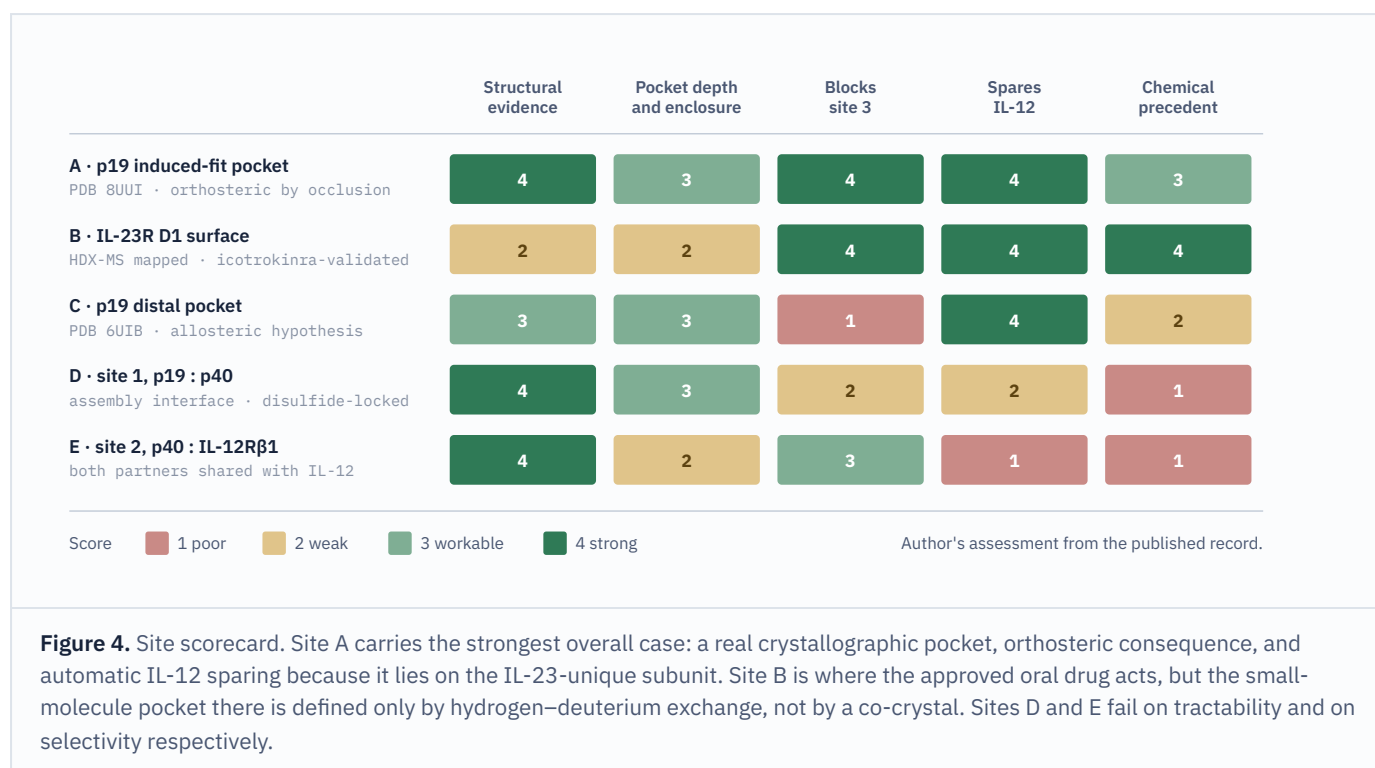


Table 4. Site-by-site assessment with the specific experiment that would confirm or kill each hypothesis.

Site	Case for	Case against	Decisive experiment
A · induced-fit pocket on p19 lead hypothesis	Crystallographically defined; lined by a defined aromatic box (Phe163), a cation- π partner (Arg162) and two backbone H-bond donors; optimised peptides in the same site reach low-nanomolar affinity with matching inhibition of IL-23R binding; a validated FP probe and counter-screen already exist	Cryptic — absent in apo structures; the reorganisation energy is paid by the ligand; a fragment may be unable to open it at all; enclosure is moderate rather than deep	Mixed-solvent MD on 5MXA plus a fragment screen by SPR or ^{19}F NMR run at high concentration, followed by soaking or co-crystallisation attempts into the 8UUI crystal form. If no fragment shows a dose-dependent, competition-sensitive signal, the site is not fragment-accessible.
B · IL-23R D1 surface high value, high risk	This is the surface the approved drug engages, so mechanism and efficacy are proven; a macrocyclic dodecapeptide binds D1 with K_D 1.81 nM and its footprint was localised by HDX-MS; binding the receptor rather than the ligand avoids competing with circulating cytokine	No public co-crystal of any ligand with IL-23R; Ig domains are typically flat; the peptide chemotype may exploit a surface groove that a 400-dalton compound cannot reach; receptor is membrane-anchored, complicating biophysics	Produce IL-23R D1 (or D1–D2) as a crystallisable construct and attempt fragment soaking with a co-crystallised known peptide as a positive control. In parallel, run HDX-MS or native MS on any hit to confirm it protects the same D1 peptides.
C · distal pocket on p19 (6UIB) optionality	A second, structurally characterised, non-overlapping pocket on the same cytokine, with a ≈ 19 nM peptide probe; could act allosterically or as a linkage point for a bivalent design	Not shown to block receptor engagement; occupying it may be pharmacologically silent; currently serves mainly as a promiscuity counter-screen	Test whether a saturating concentration of the 23–652 chemotype inhibits IL-23-induced pSTAT3 in a cellular assay. Silent occupancy means the site is only useful for counter-screening and linkage.
D · site 1, p19 : p40 deprioritise	Well characterised, including the "arginine pocket" on p40 that receives a p19 helix-D arginine; blocking assembly would prevent mature IL-23 from being secreted at all	Covalently locked by the Cys73–Cys199 disulfide once formed; assembly happens intracellularly in IL-23-producing cells, so a systemic antagonist arrives too late; the p40 pocket is shared with the IL-12 p35 interaction, so selectivity is not automatic	Only worth revisiting if an intracellular, myeloid-distributed compound is desired. Otherwise closed.
E · site 2, p40 : IL-12Rβ1 reject	Structurally well defined and known to be engineerable, as shown by the design of IL-12 partial agonists at this interface	Both partners are shared with IL-12; a blocker would recapitulate ustekinumab-like dual blockade and forfeit the principal safety and efficacy advantage of p19 selectivity	None required. Reject on target-selection grounds.

A note on what is deliberately out of scope. Downstream nodes — TYK2 (deucravacitinib and successors), JAK, ROR γ t inverse agonists — are far more chemically tractable and are already crowded. They are not this target. Similarly, targeted degradation of IL-23R is conceptually attractive but IL-23R is a type-I membrane protein, so the tractable degradation routes (LYTAC, AbTAC, bispecific receptor internalisers) are biologics, not small molecules. This dossier stays with direct, reversible, orally dosed antagonism of the p19 : IL-23R interaction.

Finding matter for a pocket that has to be persuaded to exist

The screening problem here is unusual and specific. The target conformation is a minority species; the readout must therefore be conformation-selective, and the primary screen should not be a naïve binding assay against apo protein. Three routes are worth running in parallel, with a fourth held in reserve.

Route 1 — Probe-displacement screening against the open state

The most direct approach exploits work already published. An SAR-optimised derivative of peptide 23–446 with low-nanomolar affinity has been converted into a fluorescence-polarisation probe, and a counter-screen probe based on the distal 23–652 peptide (PDB 6UIB) discriminates site-specific binders from promiscuous ones. A pilot fragment screen against this pair has been reported with acceptable assay statistics. This is a ready-made, conformation-selective primary screen: any compound that displaces the probe must be capable of stabilising or accessing the open pocket, because that is what the probe itself occupies. Running a full diversity collection and a fragment set through this cascade is the single highest-value first experiment for a new programme.

Route 2 — Peptide downsizing

Two independent macrocyclic peptide series with defined binding sites now exist: the 23–446 series on p19 and the icotrokinra/compound-C series on IL-23R. Downsizing a peptide to a small molecule is a known if difficult discipline, and here the structural information needed to do it rationally is unusually complete on the p19 side. The peptide's contacts reduce to a compact pharmacophore — two aromatics stacking on **Phe163**, an indole making a cation- π contact with **Arg162**, and two backbone hydrogen bonds to **Ser114** and **Leu116** — which is within reach of a rigid tricyclic or biaryl-amide scaffold of 380–480 daltons.

Route 3 — DNA-encoded libraries against both partners

DEL is well suited to shallow interfaces because it samples billions of compounds and tolerates weak binding. Two selections should be run: IL-23 heterodimer in the presence of a sub-saturating concentration of the open-state peptide (to bias toward the induced conformation, then compete it off in a secondary selection), and IL-23R ectodomain with and without soluble IL-23 as a competitive selection pressure. DEL's weakness — no structural information from the selection itself — is mitigated here by the availability of crystal systems for the p19 target.

Route 4 — Generative and physics-based design, held as a hypothesis generator

Published work has applied a fragment-embedding generative model to design p19-targeting scaffolds, followed by virtual screening and cellular testing in an IL-23 reporter line, with molecular dynamics used to rationalise binding at the interface. Reports of this kind are hypothesis-generating rather than confirmatory: without a co-structure or a biophysical binding constant, cellular activity in a reporter line is compatible with several off-target explanations. Treat computationally derived hits as candidates for the orthogonal validation cascade in §7, not as leads.

Table 5. Comparison of hit-finding routes, with the specific failure mode each one must be designed against.

Route	Target form	Throughput / cost	Principal failure mode	Mitigation built into the design
FP probe displacement	IL-23 heterodimer, open state stabilised by probe	10^5 – 10^6 compounds; low cost per well	Aggregators and fluorescence interference dominate the hit list	Run the 23–652 counter-screen probe in parallel; add detergent; confirm all hits by SPR and by a dose–response with varied probe concentration
Fragment screening	IL-23 heterodimer; ideally the 8UUI crystal form	1 – 2×10^3 fragments; medium cost, high information	Fragments too small to pay the loop-opening penalty; no binding detected at all	Screen at 1–5 mM with ^{19}F or STD-NMR as well as SPR; accept ligand-efficiency-driven hits with millimolar K_D ; attempt soaking into pre-opened crystals
Peptide downsizing	Both p19 and IL-23R chemotypes	Small, iterative, design-led	Loss of the macrocycle's pre-organisation collapses affinity by orders of magnitude	Retain a conformational constraint (macrolactam, atropisomeric biaryl, or fused ring) rather than going fully acyclic in one step
DNA-encoded library	Biotinylated IL-23 and IL-23R ectodomain	10^9 + compounds; front-loaded cost	Enrichment on the tag, the bead or a non-functional epitope	Matched-pair selections with and without competing peptide; off-DNA resynthesis and biophysical confirmation before any chemistry
Generative / virtual screening	8UUI conformation, plus MD-derived ensembles	Effectively unlimited; cheap	Docking scores rank solvent-exposed, greasy compounds that do not bind	Use only ligand-bound or MD-opened conformations; require a biophysical binding constant before any medicinal chemistry investment



Eight tactics for attacking the interface, and one to avoid

What follows is the medicinal chemistry consequence of §3 and §4: concrete structural strategies, each tied to a specific feature of the published structures.

1 • Anchor-first design on the aromatic box

Every reported ligand interaction in the induced pocket runs through two residues. **Phe163** provides a π -stacking face that the peptide occupies with two separate aromatics, and **Arg162** provides a guanidinium for a cation- π contact. A single rigid, electron-rich bicyclic or biaryl core placed across both — an indole, azaindole, benzimidazole or 2-aryl-imidazopyridine — can in principle recover with one ring what the peptide spends three residues achieving. This is the classic anchor-first tactic for shallow interfaces: establish a high-quality, high-ligand-efficiency contact with the deepest hydrophobic feature, then grow along the pocket rather than trying to cover the whole footprint at once.

2 • Pre-organisation to pay the loop-opening cost up front

Because the pocket is cryptic, part of every ligand's intrinsic affinity is consumed reorganising the **Thr110-Pro120** loop and helices 2–4. Flexible ligands pay twice: once for the protein, once for their own conformational restriction on binding. The design response is aggressive rigidification early rather than late — atropisomeric biaryls, fused bicyclics, spirocycles, and intramolecular hydrogen bonds that lock the bioactive conformation in solution. In cryptic-pocket programmes generally, rigid analogues frequently show a step change in affinity rather than the incremental gain rigidity usually delivers, because they are paying a fixed cost that flexible molecules cannot amortise.

3 • Macrocyclic downsizing into beyond-rule-of-five space

The realistic first destination is not a 350-dalton compound. It is a 600–850 dalton macrocycle with two or three N-methylated amides, one or two intramolecular hydrogen bonds, and a conformational chameleon profile — polar in water, closed and lipophilic in membrane. This class has delivered orally dosed drugs against other flat targets and is the natural halfway house between a 1900-dalton peptide and a conventional small molecule. Design against the **8UUI** geometry, retain the macrocycle for pre-organisation, and strip peptidic character (backbone amides, charged termini) stepwise while monitoring permeability at every round, not at the end.

4 • Two-pocket linkage

IL-23 carries two structurally characterised, non-overlapping peptide-binding pockets: the orthosteric induced-fit site (**8UUI**) and the distal site (**6UIB**). If a low-affinity fragment can be found in each, linking them is a legitimate route to nanomolar affinity on a target where neither site alone offers enough buried surface. The geometry needed for a linker can be measured directly from the two crystal structures. The risk is molecular weight: a linked pair will land in bRo5 space, so this tactic and tactic 3 converge.

5 • Reversible covalent engagement

Where non-covalent affinity is capped by a shallow surface, a reversible covalent warhead can supply residence time instead of binding energy. The relevant chemistry on a cytokine surface is lysine- or tyrosine-directed rather than cysteine-directed, since the available cysteines in IL-23 are engaged in structural and interchain disulfides. Aryl sulfonyl fluorides and fluorosulfates, 2-formylphenylboronic acids for lysine, and α -cyanoacrylamides where a suitable cysteine can be engineered or found in a paralogue are the accessible options. This is a second-line tactic: it should be attempted only after a non-covalent anchor with a defined binding site exists, because covalent labelling without a confirmed reversible binding mode is a well-known source of false leads.

6 • Receptor-side design on IL-23R D1

The approved oral drug binds the receptor, not the cytokine, and a macrocyclic dodecapeptide has been localised to D1 by HDX-MS with a 1.8 nM binding constant. Targeting the receptor has real pharmacological advantages: no competition with a circulating cytokine pool whose concentration rises during inflammation, and the possibility of engaging a receptor conformation rather than blocking an interface. The obstacle is structural information. The first investment on this route is not chemistry but protein engineering — a crystallisable or cryo-EM-tractable D1/D1-D2 construct, ideally with a fiducial binder, so that any hit can be placed.

7 • Conformational trapping rather than steric blockade

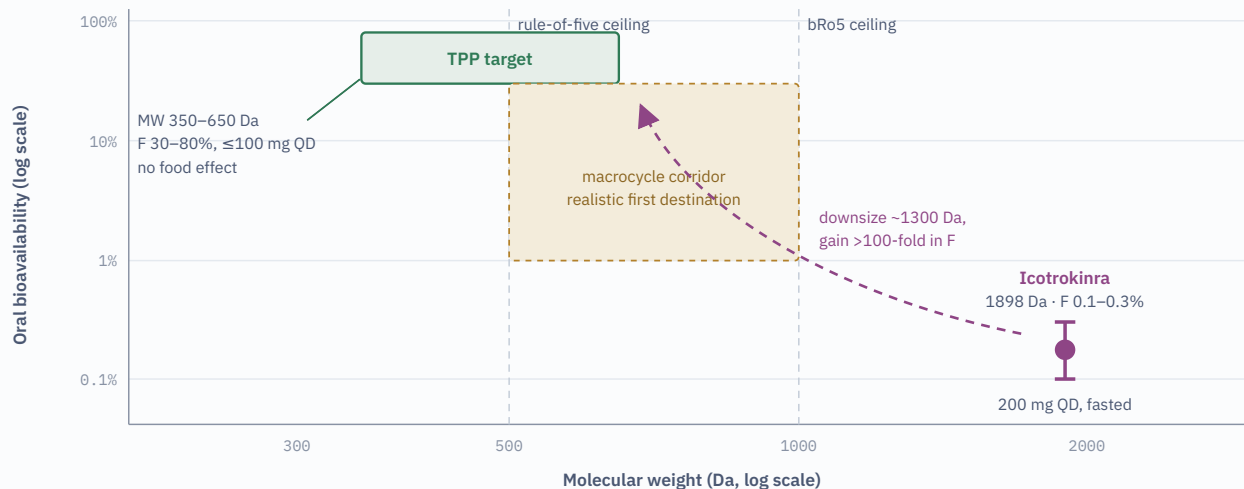
Receptor engagement partially restructures p19 and, in doing so, primes p40 to recruit IL-12R β 1 with high affinity. That cooperativity is a pharmacological handle. A ligand that locks p19 in a conformation incompatible with the productive, IL-23R-bound state would inhibit signalling without needing to bury the entire site-3 footprint — a much lower structural bar than direct competition. The 18° difference in IL-23R orientation between the binary and quaternary complexes indicates that the assembled state has real conformational latitude, which is where a modulator would act.

8 • Water-aware pocket engineering

Induced pockets of this kind are typically lined with a small number of ordered waters whose displacement dominates the enthalpic ledger. Before designing, run a grid-based hydration analysis on the **8UUI** conformation and classify each site by predicted displacement free energy. Displace only the unhappy waters; mimic the happy ones with a hydrogen-bond acceptor rather than pushing them out. On a target where total achievable affinity is constrained by surface area, getting the water treatment right is often the difference between 500 nM and 20 nM.

The tactic to avoid

Do not start with a virtual screen against apo IL-23. The pocket is absent in **5MXA**, **3DUH** and **3D87**; docking there will produce a ranked list of compounds that bind nothing, and the resulting cellular hits — when they appear — will almost always be aggregators, redox cyclers or non-specific cytotoxics. If a computational campaign is run, it must use the ligand-bound conformation or an MD-generated ensemble that demonstrably reproduces it.



Icotrokinra values are published; the target region is this dossier's proposal. Axis positions are illustrative.

Figure 6. The property gap a small molecule would close. Icotrokinra achieves clinical efficacy with sub-1% oral bioavailability by being extraordinarily potent, and pays for it with a large dose taken in a fasted state. A small molecule does not need to match picomolar potency if it can deliver two orders of magnitude more exposure per milligram; the practical route from one region to the other runs through the macrocycle corridor rather than jumping directly to rule-of-five space.

A cascade built to catch this target's specific artefacts

Two features shape the screening cascade. First, the pocket is cryptic, so binding assays must be run against a conformationally competent form of the protein. Second — and this is the constraint most likely to be underestimated — chemical matter found against the human interface may not bind the rodent orthologue. Icotrokinra happens to be cross-reactive with rat IL-23R (K_D 17.5 pM versus 7.1 pM human), which is what made conventional rodent pharmacology possible for that programme. A small molecule binding a small induced pocket has no such guarantee, and the species-coverage question should be answered in the first three months, not at candidate selection.

Table 6. Proposed tiered assay cascade. Criteria are engineering targets set by this dossier, not literature values.

Tier	Assay	Readout	Progression criterion	What it protects against
Tier 1 – binding and site				
1a	Fluorescence polarisation, optimised 23–446-derived probe on IL-23	Probe displacement, IC ₅₀	IC ₅₀ < 25 µM with a full dose–response	Nothing on its own; it is a funnel
1b	FP counter-screen with the distal 23–652 probe	Displacement at the non-overlapping site	>10-fold weaker than 1a	Promiscuous surface binders and aggregators
1c	SPR on immobilised IL-23 and, separately, IL-23R ectodomain	K _D , k _{on} /k _{off} , stoichiometry	1:1 stoichiometry; K _D confirmed within 5-fold of FP	Superstoichiometric binding, the classic aggregator signature
1d	Co-crystallography into the open crystal form; HDX-MS as fallback	Binding pose or protected peptide set	Defined site before any analogue chemistry	Optimising a scaffold that binds somewhere irrelevant
Tier 2 – cellular mechanism and selectivity				
2a	IL-23-induced pSTAT3, human PBMC or an IL-23R/IL-12Rβ1-engineered reporter line	IC ₅₀	Within 30-fold of the biochemical K _D	Cell-impermeable or serum-shifted matter
2b	IL-12-induced pSTAT4 in the same cells	IC ₅₀ ratio versus 2a	≥100-fold selective	The single most important counter-screen: loss of p19 selectivity
2c	IL-6 (gp130), IL-27, IFN-α and IL-2 signalling panel	% inhibition at 10 µM	<30%	Generic JAK/STAT or cytokine-pathway inhibition masquerading as target engagement
2d	Human whole-blood IL-23-driven IL-17A/IL-17F release	IC ₅₀ in 100% blood	<10-fold shift from 2a	Excessive plasma protein binding; sets the free-fraction target
2e	NanoBRET or fluorescent-ligand target engagement in live cells	Occupancy versus time	Occupancy tracks function	Functional activity without receptor engagement
Tier 3 – translation				
3a	Ortholog binding panel: mouse, rat, cynomolgus, human IL-23/IL-23R	K _D ratios	Within 10-fold for at least one rodent species	Discovering at candidate selection that no in vivo model is valid
3b	Rodent PK, then PK/PD with ex vivo IL-23-stimulated pSTAT3 in whole blood	Free C _{min} versus cellular IC ₉₀	≥80% pathway inhibition at trough	Dose projection built on total rather than free concentration
3c	Intradermal IL-23-induced ear inflammation; imiquimod-induced psoriasiform model	Ear thickness, epidermal thickness, IL-17A/IL-22 transcript	≥50% reduction at a tolerated oral dose	In vitro potency that does not reach the skin compartment
3d	Human skin explant or humanised model where species coverage fails	Lesional transcript signature	Directional consistency with 3c	Species-restricted binders having no in vivo evidence at all
Tier 4 – safety and developability				
4a	hERG, Nav1.5, Cav1.2; cardiomyocyte field potential	IC ₅₀ , margins	>30-fold over free C _{max}	Late-stage cardiovascular attrition
4b	Ames, in vitro micronucleus; reactive metabolite trapping	Genotoxicity, adduct formation	Negative; no structural alerts carried forward	Irrecoverable candidate failure
4c	CYP inhibition and time-dependent inhibition, induction, transporter panel	IC ₅₀ , fold induction	No TDI; reversible IC ₅₀ > 25 µM	Drug–drug interaction burden in a comorbid population

Tier	Assay	Readout	Progression criterion	What it protects against
4d	Broad secondary pharmacology panel (≥ 80 targets)	% inhibition at 10 μM	<50% at all; no GPCR/kinase hits within 100-fold	Off-target findings emerging in tox
4e	Host-defence probes: <i>Candida</i> and mycobacterial antigen recall in human PBMC	IL-17/IFN- γ recall responses	Comparable to a marketed p19 antibody at matched pathway inhibition	Over-suppression beyond the validated on-target envelope

Why 2b is the hinge of the whole cascade. The entire commercial and safety rationale for this target rests on sparing IL-12. A compound that binds p19 or IL-23R D1 gets that selectivity for free from the structure — but a compound that turns out to bind p40, or to act somewhere downstream, does not. Running IL-12-induced pSTAT4 as a matched pair in the same cells, same donor, same day, from the very first cellular experiment is the cheapest insurance in the programme.

Preclinical TPP for an oral small-molecule IL-23R : p19 antagonist

The profile below describes the compound a programme should be trying to nominate as a development candidate, not the clinical product. Every number is an engineering target chosen by this dossier to be internally consistent with a projected human dose of ≤ 100 mg once daily and with the efficacy bar set by icotrokinra. Two design decisions drive the rest: the compound must be selective over IL-12 by construction, and it must deliver at least two orders of magnitude more oral exposure per milligram than the approved peptide, since it will not match picomolar potency.

Table 7. Target product profile, preclinical / development-candidate stage. "Minimum" is the threshold below which a compound should not be nominated; "Ideal" is the profile that would support a best-in-class claim against the approved oral agent.

Attribute	Minimum acceptable	Ideal	Method and rationale
Mechanism and target definition			
Molecular target	Reversible antagonist of IL-23p19 : IL-23R site 3, binding either partner, with a structurally defined site	Co-crystal structure with the bound compound, resolution ≤ 2.5 Å	A defined site is a prerequisite for IP, for SAR and for interpreting any tox finding. HDX-MS plus mutant-shift is an acceptable substitute only transiently.
Mechanism of inhibition	Competitive or steric-occlusive with respect to IL-23R engagement; surmountable	Slow off-rate giving functional insurmountability at trough	Schild analysis against IL-23; SPR kinetics. Residence time partly substitutes for exposure.
Potency			
Biochemical affinity	$K_D \leq 100$ nM by SPR, confirmed 1:1	$K_D \leq 5$ nM	Sets the ceiling for everything downstream; must be measured on resynthesised, LC-MS-pure material.
Cellular potency	IL-23-induced pSTAT3 $IC_{50} \leq 500$ nM in human PBMC	≤ 30 nM	Primary human cells, not only an engineered reporter line, because receptor density differs by orders of magnitude.
Whole-blood potency	IL-23-driven IL-17A release $IC_{50} \leq 2$ µM in 100% human blood	≤ 200 nM	The assay that actually predicts dose. Defines the free-fraction requirement together with plasma protein binding.
Potency across IL23R genotypes	No >3-fold shift in R381 versus Q381 backgrounds	No shift	Roughly 7% of patients carry the hypomorphic allele; a genotype-dependent compound would fragment the label.
Selectivity			
IL-12 sparing	≥ 100 -fold over IL-12-induced pSTAT4, matched cells and donor	>1000-fold, and no measurable effect at 10 µM	Non-negotiable. The entire safety and differentiation case rests here.
Cytokine pathway panel	<30% inhibition at 10 µM of IL-6, IL-27, IFN-α, IL-2 signalling	<15% at 10 µM	Detects generic JAK/STAT interference presenting as target engagement.
Secondary pharmacology	<50% at 10 µM across ≥ 80 targets; no hit within 100-fold of free C_{max}	No hit >30% at 10 µM	Standard broad panel; chronic dosing in a comorbid population raises the bar.
Physicochemistry and ADME			
Molecular weight	≤ 850 Da (bRo5 acceptable with a permeability solution)	≤ 550 Da	A macrocycle in the 600–850 range is a legitimate candidate provided permeability is demonstrated, not assumed.
Lipophilicity and polarity	cLogP 1–5; TPSA ≤ 180 Å ² ; ≤ 3 hydrogen-bond donors for bRo5 chemotypes	cLogP 2–3.5; TPSA ≤ 110 Å ² ; LLE ≥ 5	Ligand-lipophilicity efficiency is the discipline that keeps a flat-interface programme from drifting into greasy, promiscuous chemical space.
Solubility	≥ 50 µM in FaSSIF; no pH-dependent precipitation across 1.2–7.4	≥ 200 µM; crystalline, non-hygroscopic, single stable form	Determines whether a food effect can be avoided — a direct differentiator against fasted peptide dosing.
Permeability	Caco-2 $P_{app} \geq 5 \times 10^{-6}$ cm/s; efflux ratio < 3	$\geq 15 \times 10^{-6}$ cm/s; efflux ratio < 2	Bidirectional with and without inhibitors; efflux liability is common in the macrocycle corridor.
Oral bioavailability	$\geq 25\%$ in rat and $\geq 20\%$ in dog or cyno	$\geq 50\%$ in both, with a food effect <1.5-fold on AUC	The core differentiation from a <1%-bioavailable peptide requiring fasted administration.
Clearance and half-life	Predicted human $t_{1/2} \geq 10$ h; CL < 40% of hepatic blood flow	$t_{1/2}$ 14–24 h supporting once-daily dosing with a flat profile	Hepatocyte intrinsic clearance across species, IVIVE, allometry with a single-species vertical scaling check.

Attribute	Minimum acceptable	Ideal	Method and rationale
Plasma protein binding	Free fraction $\geq 0.5\%$, measured, not calculated	$\geq 2\%$	Highly bound compounds in this space have repeatedly failed to translate whole-blood potency into in vivo effect.
DDI profile	Reversible CYP IC ₅₀ > 25 μM ; no time-dependent inhibition; <2-fold CYP3A4 induction	No CYP liability; not a sensitive substrate of a single enzyme	Psoriasis patients commonly take statins, antihypertensives and methotrexate.
In vivo pharmacology			
Species coverage	Within 10-fold of human affinity in at least one rodent species	Mouse, rat and cyno all within 5-fold	If coverage fails, a surrogate tool compound or a humanised system must be budgeted from the start.
Target engagement in vivo	$\geq 80\%$ inhibition of ex vivo IL-23-stimulated pSTAT3 at trough, at a tolerated dose	$\geq 90\%$ at trough with ≥ 3 -fold cover on free C _{min} /IC ₉₀	Whole-blood ex vivo assay; the bridge between PK and efficacy.
Efficacy, IL-23 ear model	$\geq 50\%$ reduction in ear-thickness increase at ≤ 30 mg/kg PO QD	$\geq 80\%$ at ≤ 10 mg/kg, matching a benchmark anti-p19 antibody	Intradermal IL-23 model; include an antibody comparator arm to define the maximum achievable effect.
Efficacy, psoriasiform model	Significant reduction in epidermal thickness and lesional IL-17A/IL-22 transcript	Effect size comparable to the antibody comparator	Imiquimod model, acknowledging its known limitations as a psoriasis surrogate.
Skin exposure	Free skin concentration $\geq$ cellular IC ₅₀ at trough	Skin:plasma unbound ratio ≥ 1	Dermal microdialysis or skin homogenate with unbound-fraction correction. A genuine potential advantage over both antibodies and a low-exposure peptide.
Preclinical safety			
Genotoxicity	Negative Ames and in vitro micronucleus	Negative, with no structural alerts in the scaffold	Standard battery run before candidate nomination, not after.
Cardiovascular	hERG IC ₅₀ > 30-fold over free C _{max} ; clean cardiomyocyte assay	hERG IC ₅₀ > 30 μM	Chronic once-daily dosing in a population with elevated cardiovascular comorbidity.
Repeat-dose tolerability	≥ 10 -fold exposure margin over the projected efficacious AUC in 14-day rodent and non-rodent studies	≥ 30 -fold, no target-organ finding	Any finding must be attributable to chemistry rather than to IL-23 blockade, given the benign class safety record.
Immune competence	<i>Candida</i> and mycobacterial antigen recall preserved at pathway-saturating concentrations, comparable to a marketed p19 antibody	Indistinguishable from vehicle	Defines whether the compound stays inside the validated on-target envelope or exceeds it.
Product concept			
Projected human dose	≤ 300 mg once daily	≤ 100 mg once daily, single tablet	Below the approved peptide's 200 mg with, critically, no fasting requirement.
Administration	Oral, once daily, food-effect <2-fold	Oral, once daily, with or without food; no titration	The primary patient-facing differentiator against current oral therapy.
Efficacy ambition	Projected week-16 PASI 90 $\geq 45\%$ and IGA 0/1 $\geq 55\%$	PASI 90 $\geq 55\%$, IGA 0/1 $\geq 65\%$, i.e. non-inferior to icotrokinra	Anything materially below the approved oral IL-23R antagonist has no path.
Cost of goods	≤ 8 synthetic steps in the longest linear sequence; no chromatographic purification at scale	≤ 6 steps; a cost structure materially below peptide manufacture	A genuine strategic advantage of small molecules over macrocyclic peptides in a price-pressured category.
Indication expansion	Mechanism supports psoriatic arthritis	Supports IBD, with a gut-exposure rationale or a locally acting back-up series	The IL-23 axis is validated in Crohn's and ulcerative colitis; a compound with high gut concentration has an obvious second life.

What kills this programme, and what it is competing against

Table 8. Risk register with explicit kill criteria. Setting these in advance is the main defence against a PPI programme consuming years on unconfirmed chemical matter.

Risk	Likelihood	Impact	Mitigation	Kill criterion
The cryptic pocket cannot be opened by molecules under 500 Da	Medium-high	Fatal	Accept bRo5 macrocycles as the target class from the outset; run fragment screening at high concentration by NMR as well as SPR; use the 8UUI crystal form for soaking	No fragment with a confirmed, site-specific K_D after two orthogonal screening campaigns
Hits are aggregators, redox cyclers or fluorescence artefacts	High	Recoverable	Detergent controls, the 6UIB counter-screen probe, SPR stoichiometry, orthogonal biophysics before any chemistry	No hit survives the Tier 1 triage from a screen of $\geq 300,000$ compounds
No rodent orthologue cross-reactivity	Medium	Costly	Run the ortholog panel at hit-to-lead, not at candidate selection; budget a surrogate series or a humanised model	Not a kill; a triggered decision point on a second series
Achievable affinity plateaus in the high-nanomolar range	Medium	Costly	Two-pocket linkage; reversible covalent engagement for residence time; accept a higher dose if the free-exposure arithmetic still supports once-daily dosing	Whole-blood IC_{50} above $2 \mu M$ with no structural route to improvement after a full lead-optimisation cycle
Efficacy falls short of the approved oral peptide	Medium	Commercially fatal	Build the PK/PD bridge early; set the phase 1 dose to achieve $\geq 90\%$ trough pathway inhibition rather than to a tolerability ceiling	Projected week-16 PASI 90 below 45% on a validated exposure–response model
Loss of IL-12 selectivity in an optimised series	Low	Fatal	Structural site selection makes this unlikely; run pSTAT4 as a matched pair from the first cellular experiment	Selectivity below 100-fold that cannot be engineered out
Freedom to operate around existing peptide and antibody estates	Medium	Manageable	Composition-of-matter on a genuinely non-peptidic scaffold; early landscape search around p19 pocket binders and IL-23R D1 ligands	Not a kill; shapes scaffold selection
Chemistry-driven toxicity in a chronic, otherwise-healthy population	Medium	Fatal	Front-load genotoxicity and secondary pharmacology; maintain LLE discipline to avoid promiscuous lipophilic chemotypes	Any genotoxicity signal not attributable to a removable structural alert

Table 9. Competitive benchmarks in moderate-to-severe plaque psoriasis. Efficacy figures are drawn from separate trials except where a head-to-head is noted, and cross-trial comparison should be treated as indicative only.

Agent	Modality and target	Route	Week-16 efficacy	Status and relevance
Icotrokinra (Icotyde)	13-residue macrocyclic peptide; IL-23R	Oral, 200 mg QD, fasted 30 min before food	IGA 0/1 approximately 65–70%; PASI 90 approximately 50–57%; superior to placebo and to deucravacitinib on coprimary endpoints	FDA approved March 2026, adults and ≥ 12 y ≥ 40 kg. The benchmark this programme must beat or match on convenience.
Risankizumab	IgG1 monoclonal; IL-23p19	Subcutaneous, every 12 weeks	Among the highest PASI 90 rates in the class	Approved. Defines the efficacy ceiling any oral agent is measured against.
Guselkumab	IgG1 λ monoclonal; IL-23p19	Subcutaneous, every 8 weeks	High and durable PASI 90; strong real-world persistence	Approved. First selective p19 antibody.
Tildrakizumab	IgG1 κ monoclonal; IL-23p19	Subcutaneous, every 12 weeks	Comparable long-term persistence with greater dosing flexibility	Approved.
Ustekinumab	IgG1 κ monoclonal; shared p40	Subcutaneous, every 12 weeks	Effective but generally below selective p19 agents	Approved. Illustrates why p19 selectivity, not p40 blockade, is the design goal.
Deucravacitinib	Small molecule; allosteric TYK2	Oral, 6 mg QD	Clearly inferior to icotrokinra on IGA 0/1 and PASI 90 in the ICONIC-ADVANCE head-to-head	Approved. The cautionary case: an excellent oral molecule that acts on the wrong node to reach biologic-level clearance.
Apremilast	Small molecule; PDE4	Oral, twice daily	Substantially lower response rates than any IL-23-directed agent	Approved. Now the floor rather than the comparator.
This TPP	Small molecule; IL-23p19 : IL-23R site 3	Oral, ≤ 100 mg QD, food-independent	Target: PASI 90 $\geq 55\%$, IGA 0/1 $\geq 65\%$	Preclinical concept. Differentiation rests on dosing convenience, cost of goods and tissue exposure, not on a novel mechanism.

The strategic read

This is no longer a first-in-class opportunity; it is a best-in-class or best-in-convenience opportunity. The mechanism has been proven orally in humans by someone else, which cuts both ways: the biology risk is close to zero, and the efficacy bar is set at the level of an injectable biologic. A programme should be justified on the specific advantages a small molecule brings over a macrocyclic peptide — food-independent dosing, an order-of-magnitude lower cost of goods, tissue and gut compartment access, and combination potential — rather than on the general appeal of the IL-23 axis, which is now fully priced in.

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Compiled 9 September 2026 from publicly available literature, structural databases, regulatory announcements and clinical trial records. Structural facts, affinities, clinical endpoints and approval status are sourced as cited. All target product profile values, assay progression criteria, cascade gates, risk scores and site scorecard ratings are the author's proposals for a hypothetical programme and carry no external authority. Figures 1–6 are schematic abstractions constructed for this document; they are not renderings of deposited coordinates. This document is a technical assessment and is not clinical, regulatory or investment advice.